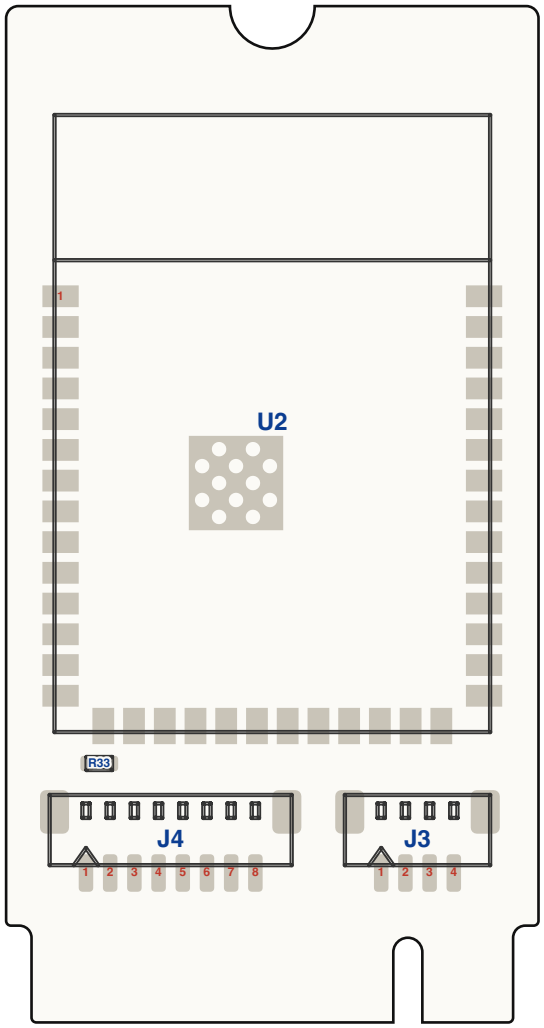
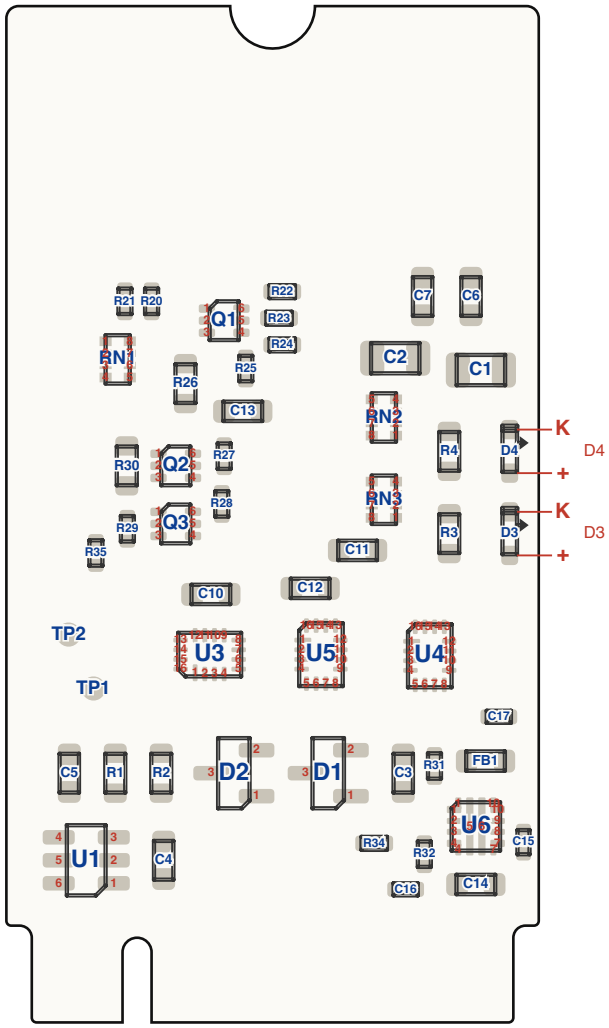
TOP



BOTTOM — seen from below



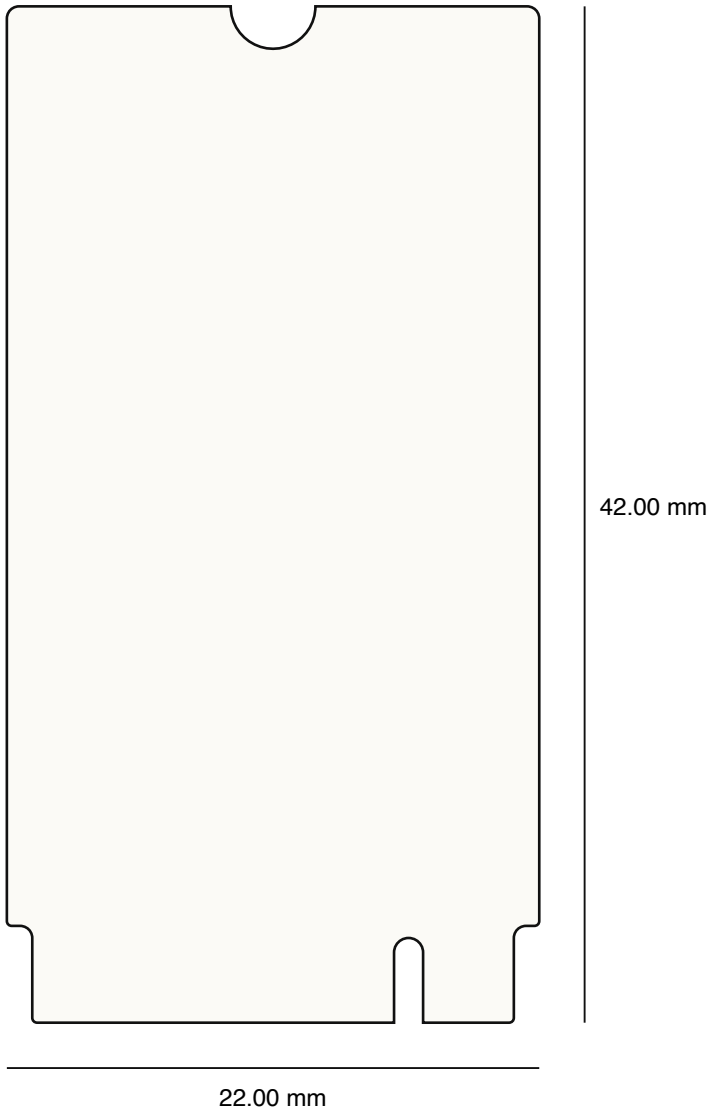
Scale 3.2:1. Grey = copper pads; bold = refdes.
Red numbers = pad numbers (polarized parts, connectors).
LEDs: '+' = anode (pad 2, wired to R3/R4),
'K' = cathode (pad 1) = SILK BAR end.
D1/D2 BAT54A: pin 3 = common anode (card side).
U1 USBLC6: pin 1 = I/O1 (D+ in), silk triangle.
J3 pin 1 = GND (Qwiic); J4 pin 1 = TXD0.
U3-U5 pin 1 = SEL1, U6 pin 1 = EN/UVLO (silk dot);
U1/U3-U6 lands are 180-symmetric: check pin 1.
Q1-Q3 pin 1 = S1 (silk triangle; 180 = equivalent).
DNP (red X): none.
TP1 TP2: bare pads, not placed (silk: function).
Bottom view mirrored like JLC's bottom preview:
N/S (screw / fingers) is invariant, E/W swap.
Verify every polarized part against the JLC
preview with fab/polarity_checklist.md.

ESP32-M2-COMPANION — assembly

Sheet 1/2 · assembly drawing

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Hole table

PTH Ø0.30 mm × 12
PTH via Ø0.20 mm × 142
PTH via Ø0.30 mm × 113

6 layers SIG/GND/SIG/SIG/PWR/SIG, 0.80 mm, JLC06081H-2116 (board: 6 Cu)
ENIG; hard-gold fingers; NO bevel (board < 50 mm):
  hand-chamfer the leading edge before first insertion
Mounting notch: routed in the outline - no drill,
  no plating: do NOT select castellated holes
Min track/space 0.127 mm (5 mil), min drill 0.20 mm
Vias (255): 113 x 0.45/0.30, 50 x 0.45/0.20, 92 x 0.35/0.20
  (pad/drill mm; 0.35/0.2 = JLC 0.2mm/(0.3/0.35mm) option)
Vias: Epoxy Filled & Capped (via-in-pad) - NOT tented
Assembly: both sides; 54 placed parts; DNP none
  top: J3 J4 R33 U2
CPL rotations: gen_fab.py JLC_ROT (bottom = library +180);
  7 offsets not yet verified in JLC's final DFM
  - VERIFY with fab/polarity_checklist.md

ESP32-M2-COMPANION — fabrication

Sheet 2/2 · fab drawing

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